

AGE AND PLAYER PERFORMANCE IN AFL SEASON 2021

A Training Report for the North Melbourne Football
Club

Abstract

A training report investigating how player performance varies with games played during the 2021 season for three different age demographics, developed in conjunction with Champion Data at the request of the North Melbourne Football Club

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Introduction

Elite athletes are getting older. It is known that players can give their best performance levels for longer now than they could a few decades ago. Training has become more sophisticated, players take better care of themselves and science has helped to understand what individuals must do to. At the elite AFL level, there are a myriad of off-field roles designed to improve player's on-field performances. Positions such as high performance, rehab and strength coaches provide necessary information and instruction designed to optimize player output in games.

As the Australian Football League 2022 pre-season approaches, there is a lot of talk in the media about "list management" by clubs. This generally involves retiring or trading "older" players. In his 2018 book *'Play On: The New Science of Elite Performance at Any Age'*, Bercovici introduces the notion of 'peak age'. This is the supposed period in which athletes are at their most likely to succeed. However, we know much less about the age at which AFL players reach their peak performance level, or about how much and in what way their performance level decreases from then on. Few studies have looked at the relationship between performance and age in the AFL.

This training report is in response to a request from the client, North Melbourne Football Club, who are looking to overhaul their player's high-performance pre-season training program for season 2022 after their disastrous 2021 campaign. Ultimately, with the aim of the new program being tailored to better suit the various age demographics of their players to yield more consistent high-level player output.

Australian Rules Football (AFL) is renowned as one of the world's toughest and most physically demanding sports. This is due to the immense running power required to play any position and uniquely, players are vulnerable to significant contact from all angles. Professional players possess immense aerobic capacities and commonly run more than 10 kilometres per game at maximum intensity efforts. Thus, due to compounding injuries and fatigue as well as the natural ageing process, this investigation will examine the effect of age on in-season player performance. In order to effectively compare and investigate player performance, the official AFL player ratings, derived in conjunction with official statistical agency, *Champion Data*, were utilised to gauge the performance of players during the 2021 season. By means of a numerical method developed through *Octave*, the respective performance of three specific AFL players in three differing age demographics being young, 'peak' and old, were analysed and investigated. The three players selected to represent each of the respective delineated age demographics were Caleb Serong (19 years of age), Ollie Wines (27 years of age) and Joel Selwood (32 years of age). The use of 3 players in these varying age brackets, allowed a comprehensive range of results to be obtained. Therefore, the following question is posed:

For 3 differing age demographics, how does player performance vary with games played during the 2021 AFL season?

Motivation & Background

In most sports, there is an age “sweet spot,” at which the combination of physical, technical, and strategic abilities comes together to produce consistent, high-level performance. Consistent, high-level performance is considered the ‘Holy Grail’ because it’s the ultimate sport and performance outcome goal. Across all sports and levels, it is common for great performances to be followed by relatively poor ones. This has generally left participants and onlookers perplexed and pondering questions. Monitoring changes in performance of players over time is an important consideration for strength and conditioning professionals and sports scientists. This information is of interest to strength and conditioning coaches who are interested in improving the performance of their players for a long-term period.

Performance analysis is used within sporting organisations to support decision-making processes relating to an individual or team’s performance. In many professional sports, various rating systems have been proposed with the aim of encapsulating player or team performance on a quantitative scale. In individual sports, the match result can be used as an objective outcome to directly compare against performance. However, rating individuals within ‘invasion’ team sports such as Australian Rules football (AFL) is more complex.

Within the competitions of many invasion team-based sports, an increase in the collection and reporting of performance data has led to the existence of more detailed and comprehensive performance rating systems. This has in turn resulted in those responsible for making organisational decisions become more reliant on performance data to make inferences about player performance and support their decision-making processes. Within the elite competition of AFL, various subjective rating systems have been proposed that quantify an individual’s match performance. However, these are susceptible to biases, which are known to accompany such subjective analyses.

Recently, a new alternative to the abovementioned systems has been proposed; the ‘AFL Player Ratings’ (<http://www.afl.com.au/stats/player-ratings/ratings-hub>). Produced by Champion Data, it is an objective system based on the principle of field equity, where a player’s actions are quantified relative to how much their actions increase or decrease their team’s expected value of the next score. This expected value is based on contextual information relating to each possession and is determined by the outcomes from every possession collected from all AFL matches preceding back to the 2004 season. Using a weighted sum of a player’s single-game rating, the Rating System gives a perpetual score that adjusts over time. A player’s most recent 30 games are given maximum weight in determining their rating, while games 31 through 40 are progressively scaled between 100% and 10%. As such, there is a buffer of at least four games where a player can miss through injury or suspension without being penalised with a reduced score.

Put simply, the rating system is all about the players who are consistently improving the position of the team, they’re the ones that are highly rewarded. The quality of a player’s involvements is considered more important than other rating systems which place more emphasis on the quantity of a player’s involvements in a game. Jackson (2016), my colleague from Champion data, discovered that the average score recorded by players is 9.5 points, with 95% of players scoring below 20.0 points. while the median is 9.0. Nearly 98% of games are above zero points and more than 95% are below 20 points, so (0, 20) acts as a natural bound for expected performance. The lowest recorded score is -8.3 in 2013; the highest is 50.5 points. There is still a heavy bias towards midfielders at the top end of the AFL Player Ratings system, but this is consistent with nearly all player awards.

Method

The method of differentiation was identified to address and solve the problem as it gives a measure of the rate at which a quantity changes. Rates of change of quantities appear in many disciplines, not just in physics and engineering.

The underlying analytical expression of the function to be differentiated is unknown. Therefore, the derivative cannot be determined analytically to solve the problem being posed. When analytical differentiation of the expression is difficult or not possible, numerical differentiation has to be used. Numerical methods can solve real world problems; however, analytical solutions solve ideal problems which in many cases do not exist in reality. A numerical solution is an approximate numerical value for the solution. Although numerical solutions are an approximation, they can be very accurate. Numerical differentiation is carried out on a function's data that are specified as a set of discrete points (tabulated data).

Three players were to be selected for analysis and investigation, allowing for obtainment of a comprehensive range of results. These three players are all to belong to three different age demographics being young, 'peak' and old, were analysed and investigated. During the selection process of the three players, the following selection bias sources were considered and subsequently avoided when choosing the three players:

- Players that are either overly young or overly old compared to most others in the AFL. Such players are overly likely to be among the very best (which would result in over-estimation of performance for all players of that age).
- Players who are missing several seasons within a typical player-career. Such players are more likely not to be among the best (resulting in under-estimation).

The three players selected to represent each of the respective age demographics were Caleb Serong (19 years of age), Ollie Wines (27 years of age) and Joel Selwood (32 years of age). These three players all play in the same position of midfield who play permanently on ball. Following the selection of these three players, to compare the players of the three different age demographic's performance across the season, the abovementioned official AFL Player Ratings compiled by Champion Data was identified as the statistic that appeared to adequately measure performance. Retrieved from the Official AFL website, the Player Ratings for each of the three selected players across their games played in the 2021 was tabulated into a CSV file, creating three different data sets.

Not all functions are linear so not all functions have constant rates of change. For a function that is nonlinear the rate of change of the function varies and may in fact be different for every different point on the graph representing the function. To fully analyse how the function is behaving, the rate of change of the function at every given point needs to be established for each of the three different tabulated data sets. For the tabulated set of points, one approach to calculate a numerical approximation of the derivative at one of the points is to use a finite difference approximation. The simple formula approximates the derivative as the slope of the line that connects two adjacent points. Once this method of numerical differentiation was selected, it was implemented into Octave Online by writing the necessary code to solve the problem. The implementation consists of a single developed m-file, which is a list of commands that allows the computer to execute the algorithm to find the solution to the problem being addressed. The derivatives at the first and last points for each individual player were calculated by using the forward and backward finite difference formula, respectively. The derivative at all the other points for each individual player was calculated by the central finite difference formula.

Results

Results for Caleb Serong	
Games Played During Season	Moving Average Rate of Change (Rating Points per Game)
1	-0.463636
2	-0.600000
3	-0.630769
4	-0.475000
5	-0.113333
6	-0.200000
7	-0.244118
8	-0.141667
9	-0.147368
10	-0.045000
11	-0.035714
12	0.102273
13	0.188095
14	0.322500
15	0.415789
16	0.438889
17	0.335294
18	0.456250
19	0.543333
20	0.485714
21	0.546154
22	0.541667

Table 1: Moving Average Rate of Change of Player Rating Values throughout Games in Season 2021 for Player in Young Age Demographic

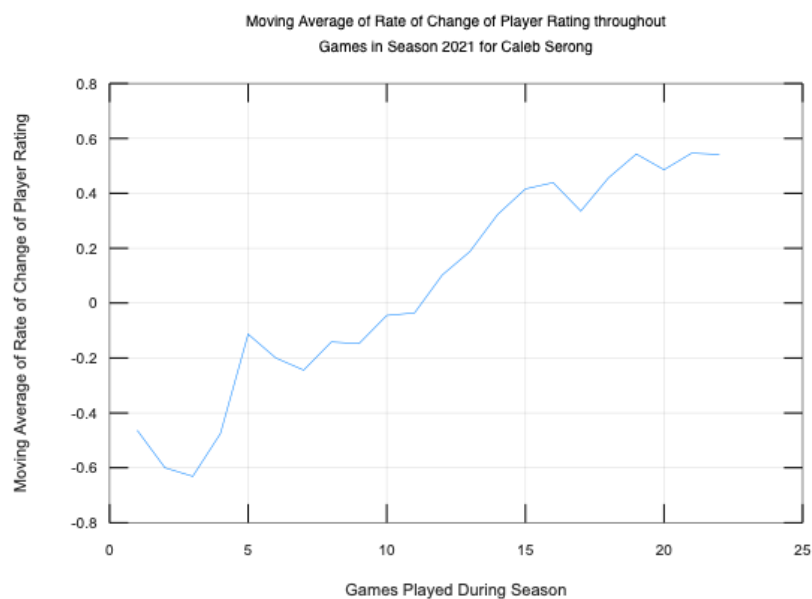


Figure 2: Plot for Moving Average Rate of Change of Player Rating throughout Games in Season 2021 for Player in Young Age Demographic

Results for Ollie Wines	
Games Played During Season	Moving Average Rate of Change (Rating Points per Game)
1	1.5045
2	1.1125
3	0.7654
4	0.8036
5	0.7867
6	0.9469
7	0.9735
8	0.7833
9	0.5842
10	0.4125
11	0.5714
12	0.8227
13	0.5429
14	0.4450
15	0.5000
16	0.6306
17	0.6588
18	0.8000
19	0.9933
20	0.7250
21	0.6769
22	0.4958

Table 2: Moving Average Rate of Change of Player Rating Values throughout Games in Season 2021 for Player in Peak Age Demographic

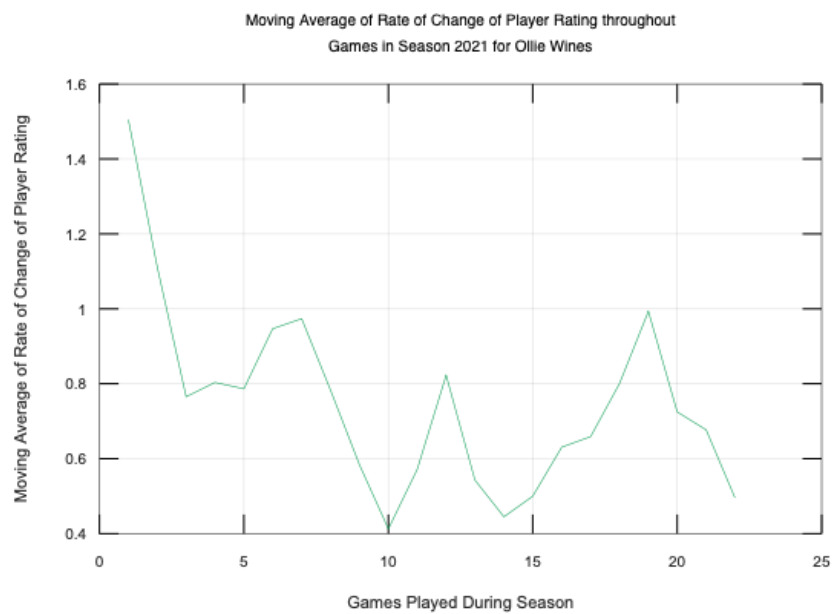


Figure 2: Plot for Moving Average Rate of Change of Player Rating throughout Games in Season 2021 for Player in Peak Age Demographic

Results for Joel Selwood	
Games Played Season	Moving Average Rate of Change (Rating Points per Game)
1	3.0000
2	-2.0000
3	-7.9500
4	-2.3000
5	6.6000
6	4.2000
7	-2.5000
8	-4.8000
9	2.2500
10	1.9000
11	-0.7000
12	4.0000
13	-4.9500
14	-3.1500
15	3.6000
16	-6.0500
17	0.3000
18	7.0000
19	-2.4000
20	-1.6500
21	-0.7000
22	-4.2000

Table 3: Moving Average Rate of Change of Player Rating Values throughout Games in Season 2021 for Player in Old Age Demographic

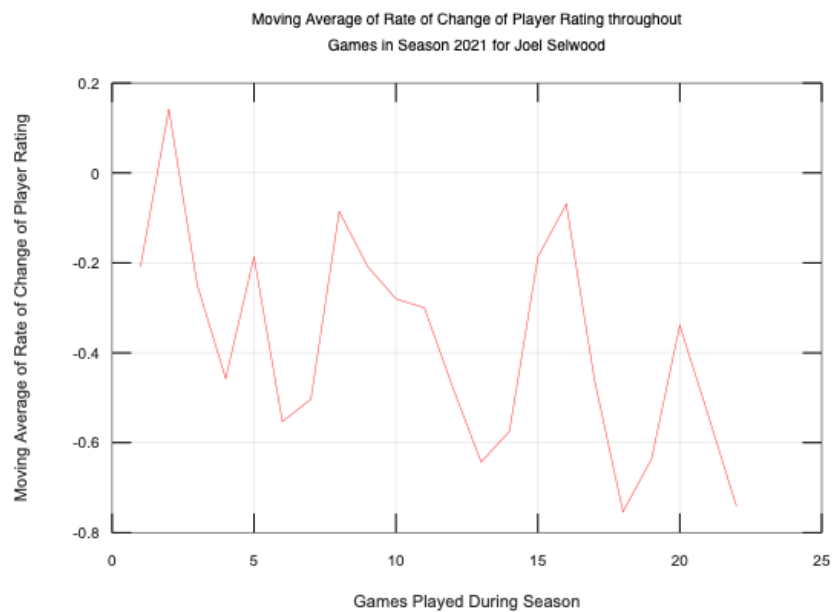


Figure 3: Plot for Moving Average Rate of Change of Player Rating throughout Games in Season 2021 for Player in Old Age Demographic

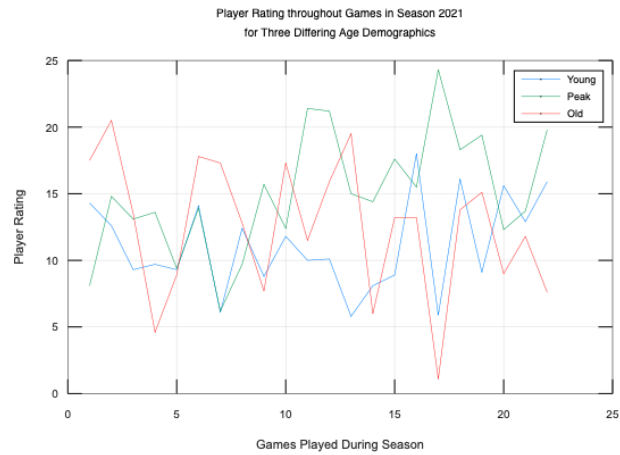


Figure 4: Plot for Player Rating throughout Games in Season 2021 for the Three Players in Differing Age Demographics

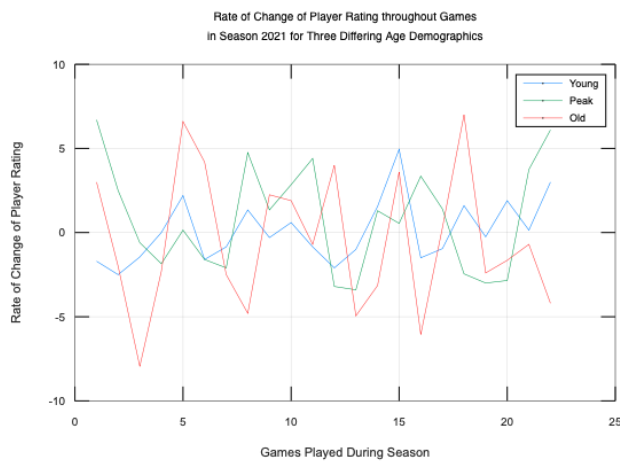


Figure 5: Plot for Rate of Change of Player Rating throughout Games in Season 2021 for the Three Players in Differing Age Demographics

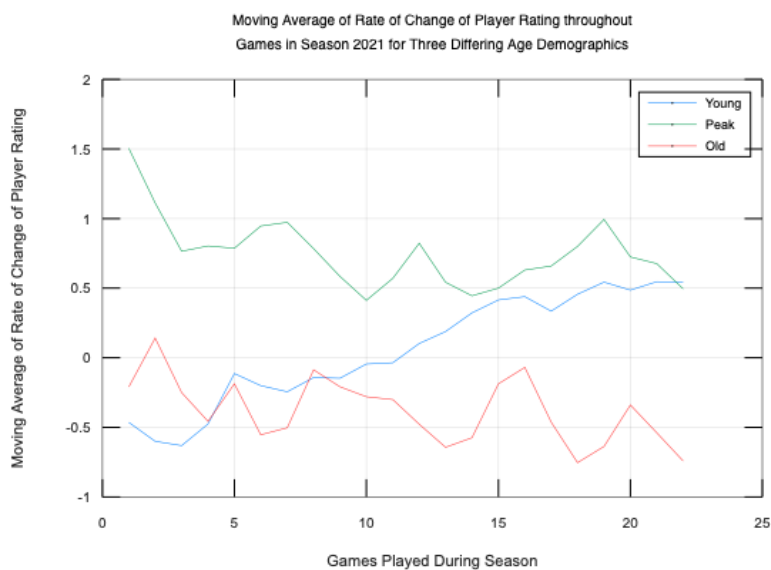


Figure 6: Plot for Moving Average Rate of Change of Player Rating throughout Games in Season 2021 for the Three Players in Differing Age Demographics

Discussion

Analysis

The initial tabulated data and rate of change results have high variability (Refer to Figures 4 & 5) which could be attributed to unknown sources, often termed 'error'. These errors are all sources of variation which the imperfect knowledge of the process cannot account for. Sources of error may include, but is not limited to, player size, such as height, hand span, arm length, body mass & skinfolds, which are all important factors in football player performance.

To better analyse the data and reveal underlying trends, the simple but very effective method of a moving average was implemented to yield satisfactory results that are as close as possible to the real data. This process aids in the simple visualisation of the data. The order of the moving average, or in other words the window size, determines the smoothness of the curve. The wider the window size the smoother the result (lower variance), but the less responsive it will be (which is where the greater bias comes in), while narrower windows will have less bias but greater variance. The averaging window is moved over the data, shifting it by one time step after each calculation. This smooths the data and removes random variations. The smoothing process is also referred to as filtering the data. A window size of 22 was chosen as there are 22 rounds in the AFL season and yielded satisfactory underlying trends.

Analysis of the data and ensuing graphs developed by the numerical method in Octave revealed the following findings. Firstly, for player representing the young age demographic, Caleb Serong, Figures 1 & 6 indicated a steady and proportional increase in performance in comparison to games played. Therefore, it can be said that the rate of change of his player performance with respect to games played is positive for the domain of the 2021 season. As the function is increasing somewhat linearly for the 2021 season domain, it has a constant rate of change and thus the rate of change is simply the gradient of the graph. For the player representing the peak age demographic, Ollie Wines, Figures 2 & 6 demonstrate that his particular performance across the 2021 season remains at a relatively constant and high-level. Therefore, as player performance remains similar as games played changes, the rate of change of his player performance with respect to games played is said to be near zero. Conversely, for the player representing the old age demographic, Joel Selwood, Figures 3 & 6 reveal significant fluctuations in performance between poor and mediocre across the 22-game sample size. Therefore, as established earlier, the rate of change of the function is in fact different at each different point on the graph of the function.

As they age, many athletes complain that the ability to recover from hard bouts of exercise diminishes. For athletes, experiencing more training-associated injuries leads to reduced training intensity and volume, and thus poorer performance come game day. The results of Joel Selwood, the player from the old age demographic, confirm this.

A publication titled '*Applied Sport Science of Australian Football: A Systemic Review*', conducted at Brisbane's Australian Catholic University in 2018 by three academics and one AFL club sport scientist employee, allowed for a multitude of conclusions to arise that are relevant to the question being investigated in this training report and assist in understanding and explaining the results obtained from the numerical method. Firstly, it was found that the physical demands of AFL football exceeded those of other sports due to the length and nature of the game. Secondly, elite footballers must be given personalised, tailored training regimes that consider their age, body shape and role in the game in an effort to build tactical and physical skills. In conjunction with this, internal and external workloads of players must be managed to not only improve performance, but also minimise potential injuries (Johnson et al., 2018).

The advantages of using the Player Rating System as a measure of performance for this training report is that by using nearly all available events within the game and assessing them based on the mathematical value of each event, a good representation of a player's contribution is obtained. The disadvantages of the Player Rating System as a measure of player performance for this report is that it is deficient in ranking

defensive minded player positions as they have limited on-ball activity, which affects the ability of the ranking system to fully capture their involvement. Fans and industry people are quick to complain over how often these defenders are overlooked and discriminated against using purely statistical measures, however even subjective assessments of players such as various awards are rarely kind to defenders and heavily favour midfielders. Moreover, the system was merely available from 2010 and since two seasons of games are included in the AFL Player Ratings, overall standings are only available from the start of the 2012 season. A further disadvantage of the Player Rating System is it is based on the players most-recent 40 games within a two-season block, with a restriction that they were played within the previous two-year period. However, though this may be interpreted as a weakness of the Rating System, it could also be interpreted as a strength. Players are slower to reach their true standing after a rise in form, but it ensures that overall standings are based on repeated, consistent form, as opposed to shorter bursts of form that may not be lasting.

Validity of Results

CEPAR (2012) found that the average and median age of male AFL players increased from 23 in 1967 to 25 in 1983. This period coincided with growing professionalisation of the sport. "The median age of male AFL players has remained relatively stable around age 25, though maximal and average ages increased slightly. The average age increased from 24 in 1991 to 25 in 2019. The stable age of 25, lower than in tennis, football, and cricket, may reflect the fact that AFL is a multi-skill sport. Speed is required for moving across the 155-170m long playing field; strength is required to *mark* the ball from the air, muscle opponents out of the way, and when tackling. The greater contact and impact means that there is a greater premium on fast recovery, which is more difficult at older ages" (CEPAR, 2012).

McIntosh et al (2019) emphasise that considerations or assumptions must be made about other factors when assessing player performance. Notably, a player's position and their level of experience. In addition to these factors, another consideration in the AFL is the position at which players are selected in their respective draft. McIntosh et al's (2019) study on longitudinal player performance in the AFL results indicate that age and experience had the largest influence on performance in each of the models, respectively, followed by positional role. The results also indicated that performance continues to improve throughout an AFL players career. However, as noted by McIntosh et al (2019), "this type of analysis is susceptible to selection biases. Specifically, previous research has identified that these biases can be bought upon as a result of better-performing players typically having longer careers than other players" (McIntosh et al., 2019).

HPN Footy (2015) further support this theory that "better-performing players typically have longer careers than other players," by combining the two effects of player improvement and player attrition. "There's roughly twice as many 21-year-olds as 27-year-olds and twice as many 27-year-olds as 31-year-olds" (HPN Footy, 2015).

In other sports such as soccer, several recent studies could provide some proof on the relationship between age and performance. The conclusions of these studies suggest significant aspects to have in mind. There is a clear loss of physical performance in players over 30 years compared to younger footballers. After analysing 10,739 players during the 2017-2018 season, researchers discovered that the loss of performance is especially drastic in those over 35. This trend was noted across all positions. However, the technical-tactical performance appears to be better in older players. It is possible that the deterioration in physical performance of younger players is compensated by an improvement in other skills such as decision making and game intelligence (Rey et al., 2019).

Recommendations and Conclusion

Many variables may affect performance, but the main purpose of this research was simply to investigate the relationship between games played during the 2021 season and player performance by comparing three players of differing age demographics. For every type of mathematical problem there are several (or many) numerical techniques that can be used. The techniques differ in accuracy, length of calculations, and difficulty in programming.

When the data to be differentiated is obtained from experimental measurements, usually there is scatter in the data because of the experimental errors or uncertainties in the measurement. If this data set is differentiated using a two-point finite difference approximation, which is the simplest form of finite difference approximation, then large variations (positive and negative values) will be seen in the value of the derivative from point to point. Better results can be obtained by using higher-order formulas of finite difference approximation that consider the values from more than two points. For example, there are four, five, and seven-point finite difference formulas. An alternate approach to differentiation of data specified by a set of discrete points is by curve fitting the data with an analytical function that can be easily differentiated. The approximate analytical expression is then differentiated for calculating the derivative at any of the points. In this case, the data is smoothed out before it is differentiated, eliminating the problem of wrongly amplified slopes between successive points. This procedure may be preferred when the data contains scatter, since the curved-fitted function smooths out the variability. However, these alternative, more involved methods, and approaches to solve the problem are beyond the scope of this training report.

Furthermore, as the maximum window size of 22 was chosen to smooth the data from these large variations and hence, better reveal trends, it is less realistic. This is because higher window size means more smooth data, and thus less realistic. Moreover, a limitation of using the AFL Player Ratings as a measure of performance is that the methodology could include additional metrics, such as time on ground or spatiotemporal data, potentially allowing for further explanation of the results. The results of McIntosh et al's study (2019) further support the notion that overall player performance evaluations should consider both subjective and objective assessments in a complementary manner to accurately evaluate player performance.

The implementation of this numerical method may provide valuable knowledge for professional AFL organisations such as North Melbourne. Implications of the use of the numerical method could assist with professional sporting organisation's operational decision-making relating to player recruitment, contracting and development. Future work should focus on the refinement of the numerical methods produced and used in this study as additional seasons of data become available. Moreover, future work should focus on the continual development of improving the ratings system as new technologies become available, as well as the interpretation and application of the AFL Player Ratings system for objective performance analysis and operational decision-making.

Variability is a natural part of the human condition and human performance. The reality is we will never know exactly what ingredients went into making up a performance. At best we may be able to develop a hunch that links some elements of effort to some variations in results, with a whole pile of unknown's leftover. In its simplest form 'a performance' is the consequence of about 25 to 30 areas of effort. On top of these areas of effort we also have many less influenceable elements aspects such as genetics. This is something that should always be carefully considered. Every player is unique. Their needs must be adapted to. Older athletes need longer to recover and adapt to a training stimulus, so program planning needs to change with age. This reduces injuries and provides players with a competitive advantage in terms of repeat high-intensity efforts and strength/power. Performance decline isn't just about physical changes, however. As we age, our intrinsic motivation to train diminishes, even in professional athletes. This should also be heavily considered...

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